

ML Foundations – Week 4 Day 1

1. Problem Framing

The goal is to predict whether a person earns **more than \$50K per year** using the Adult Income dataset. This helps marketing teams target high-income customers and reduce spending on customers who are less likely to respond.

2. Chosen Evaluation Metric

Primary Metric: Precision

Reason: Precision minimizes false positives. Predicting a low-income customer as high-income leads to unnecessary marketing costs, so high precision is important.

3. Baseline Results

Metric	Majority	Rule-Based
Accuracy	0.7607	0.7530
Precision	0.0000	0.4844
Recall	0.0000	0.4970
F1 Score	0.0000	0.4906
ROC AUC	0.5000	0.6653
PRAUC	0.2393	0.3611

Interpretation:

The Majority Baseline predicts every sample as the majority class and fails to identify high-income individuals. The Rule-Based Baseline (**education-num ≥ 13**) performs much better and serves as a stronger benchmark.

4. Error Analysis

- **False Positives:** 1,237 records (Avg. Education: **13.31**, Avg. Age: **37.23**).
- **False Negatives:** 1,176 records (Avg. Education: **9.54**, Avg. Age: **44.07**).

Education alone is not enough to predict income, as other factors such as occupation, experience, and working hours also affect earnings.

5. Next Steps

- Handle missing values.
- Encode categorical variables.
- Transform skewed numerical features.
- Train Logistic Regression and Decision Tree models.
- Continue optimizing **Precision** while improving Recall and F1 Score.